QRP VLDO V2.1

M9OMS

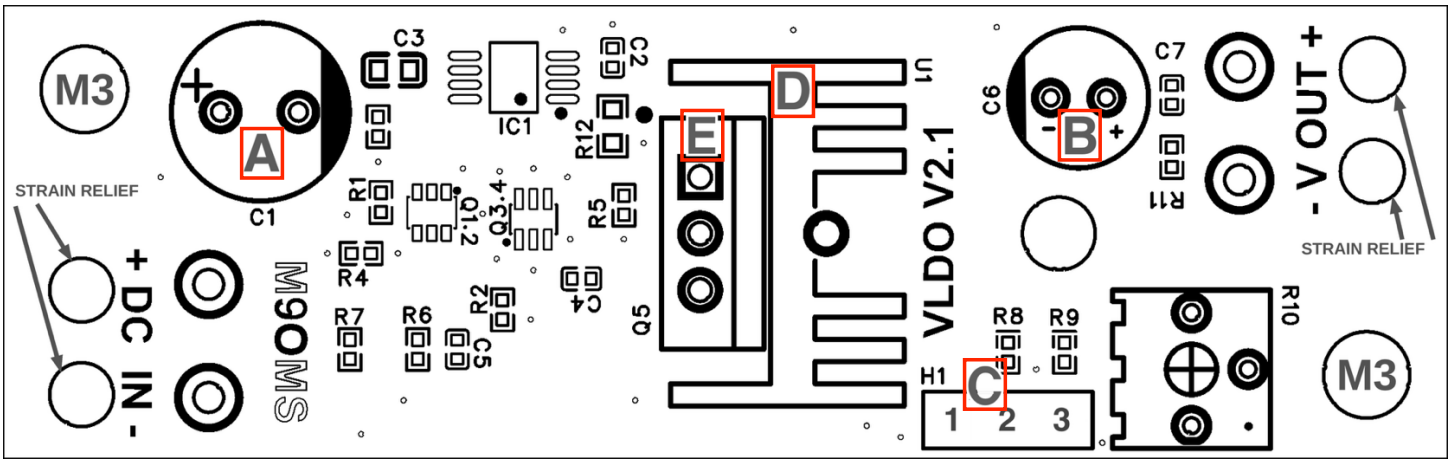


Figure 1: Top of board, with assembly steps labelled A–E.

Assembly instructions

1. **A —** Solder C1 (470µF, red electrolytic capacitor). Observe polarity — the long lead is positive.
2. **B —** Solder C6 (220µF, silver polymer capacitor). Observe polarity.
3. **C —** Solder H1 (header pins). Tack one leg down first, then reflow that joint while pressing the header flat against the board to get it square. Watch your fingers — the pins get hot. Once it sits flat and square, solder the remaining legs.

⚠ HANDLE PMOS WITH CARE — ESD-sensitive device. ⚠

Decide now how you will mount the PMOS: with the included heat sink (continue to step **D**), or bottom-mounted directly to an external case or cooling solution (skip to step **F**). **The PMOS is fully isolated — no isolating pads are necessary.** Using an isolating pad will increase thermal resistance. Use thermal paste only.

Apply a small amount of thermal paste to the back of the PMOS. Place it onto the heat sink and gently secure it with the M3 machine screw. Do not tighten fully yet — the PMOS should still be able to move from side to side, so you can ease its legs into the board pads in step D.

4. **D —** Place the PMOS (Q5) + heat sink (U1) assembly onto the top of the board. The heat sink should sit flush. Solder the heat sink pin: this joint is mechanical only and carries no current, but apply enough solder to immobilise the pin. Use a HOT iron for this. Ensure the assembly is secure, straight and square before continuing.
5. **E —** Tighten the M3 screw fully. Solder the PMOS gate pad (the square one) at normal iron temperature. Then solder the two round pads — additional heat will be required, particularly for the centre pin.
6. **F —** *Bottom-mount option (PMOS bonded directly to your case or external cooling solution).* Dry-fit the PMOS (Q5) referencing the clearance hole on the rear of the board, with its writing facing the PCB, and note where the legs need to bend. Bend them carefully to a right angle. Bond the PMOS to your cooling solution first (don't forget thermal paste), then fit the board over the legs and solder them — solder after everything is mechanically secured (including the board) for the best result. The clearance hole stays accessible from the top of the board, so you can remove the whole board without desoldering if required.

Wiring and strain relief

The input and output pads accommodate up to 18 AWG stranded wire – keep the cables short, pay close attention to polarity. Use the strain relief holes to secure your cables:

- Pass the wire through the strain relief hole from the top of the board.
- Bring it back down through the pad hole from the bottom, leaving a tight loop.
- Pull the loop snug, then solder. Any pull on the cable now acts on the loop rather than the solder joint.

A small cable tie through the strain relief holes can work as an alternative.

Selecting the output voltage

The header pins (H1) allow you to select the desired output voltage. Jumpers/shunts are included.

- **No shunt** ≃ 9 V
- **Shunt pins 1 + 2** ≃ 12 V
- **Shunt pins 2 + 3** ≃ 13.8 V

Check the output voltage is correct before connecting the board to sensitive equipment. Adjust the output (with $V_{in} > V_{out}$) using the trim potentiometer R10. If there is any subsequent drift, re-adjust R10.

Recommended operating parameters

- **Input:** 8*–18 V DC
- **Regulated output:** 9.0, 12.0, 13.8 V DC (trimmable)
- **Max current:** 2 A

** A DC supply stiff enough to keep the input voltage above this threshold is necessary.*

Heat sink and housing

The supplied heat sink should dissipate up to 1.5 W continuous, or 3 W of heat at a 50% duty cycle, provided it has sufficient ventilation.

There are various ways to house the board. Please ensure there are plenty of ventilation holes when housing it in a project box or 3D-printed case.

For large current draws, voltage drops or heavy duty cycles: it is advisable to house the board in a metal project box, with the PMOS bonded directly to the case for better heat dissipation (see step F). The Multicomp MC003550 has good dimensions and is available from CPC for under £4 – in my own tests, it dissipated 4 W continuous for 2 hours without exceeding thermal limits.

Board layout note

The bottom of the circuit board is a +DC OUT power plane (for thermal management). The top of the board is a ground plane. **A no-copper area has been provided around the M3 mounting holes, but care must be taken not to short the two sides when securing the board down.** Use plastic or cardboard washers if your metal hardware is larger than the safe area provided.

Project Credits:
KC7XE — Static DC and thermal measurements.
CR7BTQ — Dynamic measurements.
M9OMS — Schematic, circuit design, PCB layout, and prototyping.

https://radio-uncensored.github.io/qrp-vldo-regulator/design.html